

Ian Sebastian V. Macabulos

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PROJECTS

UA LabSign – University Laboratory Attendance | Next.js, TypeScript, PostgreSQL, React Native, Expo

- Engineered a web and Android attendance platform for the University of the Assumption, replacing manual paper-based laboratory attendance with centralized digital workflows designed to support hundreds of students, 20+ faculty members, and university administrators across multiple academic programs.
- Designed a secure, device-bound attendance protocol combining ECDSA P-256 signatures, GPS geofencing, institutional Google authentication, and expiring session PINs to verify user identity, authorized device, physical presence, and active class session before recording attendance.
- Architected synchronized Next.js and React Native/Expo clients backed by PostgreSQL and Prisma, supporting real-time attendance monitoring, class scheduling, report generation, device authorization and recovery, RBAC, and administrative audit trails.

AC-CORE – Municipal Maintenance and Incident Reporting System | Angular, Node.js, Express.js, MongoDB

- Engineered geospatial flood-risk routing across 14 high-risk neighborhoods using elevation data to identify potential downstream impact areas and support infrastructure prioritization.
- Developed a Paved Paradox prioritization model for road maintenance, combining infrastructure condition and commercial-area risk to surface higher-impact repairs during heavy rainfall.
- Built a responsive Angular Zoneless + Leaflet interface with marker clustering, GPS auto-detection, and a 30-second live work-order refresh for faster field reporting and monitoring.

TECHNICAL SKILLS

Frontend: Next.js, React, TypeScript, Tailwind CSS, Vue.js, Angular, React Native, Expo

Backend & Databases: Node.js, Nest.js, Express.js, FastAPI, PHP, Laravel, Java, Python, PostgreSQL, MongoDB

Tools & DevOps: Jest, Vitest, Playwright, Postman, Git, GitHub Actions, Vercel, AWS, Figma, WordPress

WORK EXPERIENCE

Backend AI Engineer Intern — FlyRank AI

- Engineered a TypeScript/Fastify usage metering and billing service for LLM applications, implementing tenant quotas, token tracking, idempotent requests, and micro-cent usage pricing with PostgreSQL.
- Integrated Stripe test-mode billing and webhook processing, with automated tests covering quota enforcement, usage accounting, checkout flows, and billing events.

Full-Stack Web Developer Intern — Sortr

- Built an internal IT service desk ticketing system for 50+ employees within the company domain, centralizing hardware and software issue reporting, automating ticket routing and status tracking, and providing the IT team with a structured support workflow from submission to resolution.
- Redesigned and optimized WordPress production pages, improving responsive layouts, content structure, technical SEO, image delivery, and overall frontend performance.

Freelance Full-Stack Developer

- Built and deployed custom full-stack systems for local businesses and academic clients, replacing manual workflows with web applications for attendance, operations, reporting, and data management.
- Managed projects end to end—from requirements gathering and system architecture to database design, authentication, testing, deployment, and client handoff—delivering solutions used by dozens to hundreds of users across different academic and business workflows.

EDUCATION

Holy Angel University

2023 – Expected Graduation: 2027

Bachelor of Science in Information Technology, Major in Web Development

- Academic Honors:** Consistent Dean's and President's Lister, A.Y. 2023-2026
- Certifications & Training:** CompTIA IT Fundamentals (ITF+), AWS Cloud Foundations (AWS Academy), freeCodeCamp (Responsive Web Design, JavaScript Algorithms, Back End Development).